👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 8 Engine Teardown & Rebuild (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Engine project checklist and grading**: The project steps and rubric are done in class and not written in Canvas; our checklist is team-written.
- **Troubleshooting**: A syllabus goal with no class materials; our lessons are team-written.
- **iCADEMY tech bits**: STIHL iCADEMY is login-only; we wrote our own service content from public knowledge.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

🔧 Engine Teardown & Rebuild: Guided Notes

Small Engines Overhaul · Unit 8 · Level 3 (closer to the class work) · goes with the engine project, Ring End Gap handout, valve video and service topics

Name: _____

Use the class slides and your reading. Key words:

teardown *flywheel puller* *compression ring* *oil control ring* *end gap* *side clearance* *lapping* *valve clearance* *timing marks* *torque* *troubleshooting*

A. Teardown

1. List the teardown order from outside parts to the crankshaft.

_____

_____

_____

B. Rings and valves

2. Fill in the table.

Check	Tool	If out of spec
Ring end gap		
Ring side clearance		
Valve clearance		

C. Troubleshooting

3. An engine has spark and compression but won't start. What do you check next and how?

_____

_____

🔧 Engine Teardown & Rebuild: Reading

Small Engines Overhaul · Unit 8 · Level 3 (closer to the class work) · goes with the engine project, Ring End Gap handout, valve video and service topics

Name: _____

Overhauling a Small Engine

An overhaul starts with a careful teardown. With adult supervision, disconnect the spark plug wire and drain the fuel and oil. Remove outside parts first: shroud, air filter, carburetor and muffler. A flywheel puller removes the flywheel; then the armature, cylinder head, valves, piston and connecting rod, and crankshaft come out. Technicians photograph each step and keep parts in a labeled tray so reassembly is easy.

During inspection, the piston rings are checked. There are two compression rings and an oil control ring. Ring end gap is measured by pushing a ring squarely into the bore and sliding a feeler gauge into the gap; side clearance is measured with the ring in its groove. Valves are cleaned, inspected and lapped with compound until an even gray band shows a good seal, then valve clearance is set to spec.

Reassembly is the reverse of teardown, using new gaskets and oil on moving parts. The timing marks on the crankshaft and camshaft gears must line up so the valves open at the right time, and head bolts are tightened in the manual's pattern to the torque spec.

When an engine won't run, technicians troubleshoot in order: fuel, spark, compression and air. The same skills apply to outdoor power equipment such as chainsaws, trimmers, blowers and cut-off saws, along with safe practices like correct chain tension, chainsaw chaps and dust control for cutting concrete.

Check your reading

1. Why photograph and tray parts during teardown?

2. How is ring end gap measured?

3. List the four troubleshooting checks in order.

🔧 Engine Teardown & Rebuild: Lab Sheet

Small Engines Overhaul · Unit 8 · Level 3 · goes with Engine project + Engine Assembly 1 + Ring End Gap Handout

Name: _____

Question: Is your engine's piston and valve condition within spec, and does the rebuilt engine run?

Materials

engine, tool set, flywheel puller, feeler gauges, torque wrench, lapping kit, parts tray, manual

⚠️ Safety

- Safety glasses and coveralls
- Plug wire disconnected; fluids drained with adult supervision
- Springs: eye protection, tools with adult supervision
- Start engine outdoors with an adult

Prediction

Predict which part will be most worn.

_____

_____

Procedure

1. Tear down in order with photos and labeled trays.
2. Measure ring end gap and side clearance.
3. Lap valves; set clearance.
4. Reassemble with timing aligned and bolts torqued.
5. Test fuel, spark, compression; run with adult.

Data

Check	Spec	Measured	Action
Top ring end gap			
Ring side clearance			
Intake valve clearance			
Exhaust valve clearance			
Head bolt torque			

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was wrong with your engine and did the rebuild fix it? Evidence: your measurements and test. Reasoning.

🔧 Engine Teardown & Rebuild: Quick Check

Small Engines Overhaul · Unit 8 · Level 3 (closer to the class work) · goes with the engine project, Ring End Gap handout, valve video and service topics

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The flywheel is removed with a…

A. Hammer | B. Torque wrench | C. Flywheel puller

2. The bottom piston ring is the…

A. Oil control ring | B. Top compression ring | C. Valve ring

3. Lapping valves produces…

A. More clearance | B. Rust protection | C. An even sealing band

4. Head bolts are tightened…

A. In pattern to torque spec | B. As tight as possible | C. One at a time fully

5. Why must timing marks line up during reassembly?

6. Describe two safety practices for chainsaw service or use.

🔧 Engine Teardown & Rebuild: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 3 (closer to the class work) · goes with the engine project, Ring End Gap handout, valve video and service topics

Name: _____

Use the class slides and your reading. Key words:

teardown | *flywheel puller* | *compression ring* | *oil control ring* | *end gap* | *side clearance* | *lapping* | *valve clearance* | *timing marks* | *torque* | *troubleshooting*

A. Teardown

1. List the teardown order from outside parts to the crankshaft.
Plug wire/fluids, shroud, air filter, carb, muffler, flywheel, armature, head, valves, piston/rod, crankshaft.

B. Rings and valves

2. Fill in the table.

Check	Tool	If out of spec
Ring end gap	**feeler gauge**	**replace rings / check bore**
Ring side clearance	**feeler gauge**	**replace piston**
Valve clearance	**feeler gauge**	**grind stem or adjust**

C. Troubleshooting

3. An engine has spark and compression but won't start. What do you check next and how?
Fuel: fresh fuel reaching the carb, clean jets, float/needle working; then air filter/choke.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🔧 Engine Teardown & Rebuild: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 3 · goes with Engine project + Engine Assembly 1 + Ring End Gap Handout

Name: _____

Question: Is your engine's piston and valve condition within spec, and does the rebuilt engine run?

Materials

engine, tool set, flywheel puller, feeler gauges, torque wrench, lapping kit, parts tray, manual

⚠️ Safety

- Safety glasses and coveralls
- Plug wire disconnected; fluids drained with adult supervision
- Springs: eye protection, tools with adult supervision
- Start engine outdoors with an adult

Prediction

Predict which part will be most worn.

Answers vary; often rings or exhaust valve.

Procedure

1. Tear down in order with photos and labeled trays.
2. Measure ring end gap and side clearance.
3. Lap valves; set clearance.
4. Reassemble with timing aligned and bolts torqued.
5. Test fuel, spark, compression; run with adult.

Data

Check	Spec	Measured	Action
Top ring end gap	**0.007-0.017**	**0.025**	**replace rings**
Ring side clearance	**0.002-0.004**	**0.003**	**OK**
Intake valve clearance	**0.005-0.007**	**0.006**	**OK**
Exhaust valve clearance	**0.007-0.009**	**0.004**	**grind stem**
Head bolt torque	**140 in-lb**	**140 in-lb**	**done**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was wrong with your engine and did the rebuild fix it? Evidence: your measurements and test. Reasoning.

Example: worn rings (large end gap) caused low compression; new rings raised compression and it started.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🔧 Engine Teardown & Rebuild: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 3 (closer to the class work) · goes with the engine project, Ring End Gap handout, valve video and service topics

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The flywheel is removed with a…

A. Hammer | B. Torque wrench | **C. Flywheel puller**

2. The bottom piston ring is the…

A. Oil control ring | B. Top compression ring | C. Valve ring

3. Lapping valves produces…

A. More clearance | B. Rust protection | **C. An even sealing band**

4. Head bolts are tightened…

A. In pattern to torque spec | B. As tight as possible | C. One at a time fully

5. Why must timing marks line up during reassembly?

So the valves open and close at the right time in the 4-stroke cycle.

6. Describe two safety practices for chainsaw service or use.

Chain brake/plug wire off, gloves, chaps, eye/ear protection, correct tension.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*